Microsoft Agent Framework and Azure AI Foundry Agent Service

Prepared By

SREEDEV C S

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Contents

1 Introduction to AI Agents **1**

2 Understanding Microsoft Agent Framework **2**
2.1 What is Microsoft Agent Framework? 2
2.2 Design Goals 3
2.3 Core Components 3
2.3.1 Agent 3
2.3.2 Planner 4
2.3.3 Memory 4
2.3.4 Tools 4
2.3.5 Models 4
2.4 Execution Workflow 4
2.5 Architecture 4
2.5.1 Agent Runtime Layer 5
2.5.2 Reasoning Layer 5
2.5.3 Planning Layer 5
2.5.4 Tool Integration Layer 5
2.5.5 Memory Management 5

3 Azure AI Foundry Agent Service **6**
3.1 Introduction 6
3.2 Key Features 7
3.3 Built-in Tools 7
3.4 Enterprise Integration 7

4 Microsoft Agent Framework vs Azure AI Foundry Agent Service **8**
4.1 Microsoft Agent Framework 8
4.2 Azure AI Foundry Agent Service 8
4.3 Comparison 9

Chapter 1

Introduction to AI Agents

Artificial Intelligence (AI) has evolved significantly from rule-based expert systems to modern Large Language Models (LLMs) capable of generating human-like responses. While traditional AI systems excel at answering questions or generating content based on user prompts, they are generally reactive and limited to single interactions. Recent advances have led to the emergence of **Agentic AI**, where AI systems can reason, plan, invoke tools, maintain memory, and perform actions toward achieving a goal. Unlike conventional chatbots, agents can independently determine the sequence of steps required to accomplish a task. Microsoft has become a major contributor to the agentic AI ecosystem through technologies such as the **Microsoft Agent Framework (MAF)** and the **Azure AI Foundry Agent Service**. These technologies enable organizations to build intelligent, scalable, and secure AI agents for enterprise environments.

Chapter 2

Understanding Microsoft Agent Framework

2.1 What is Microsoft Agent Framework?

Microsoft Agent Framework (MAF) is an open and extensible framework designed for developing intelligent AI agents capable of reasoning, planning, interacting with tools, and collaborating with other agents. The framework provides the foundational infrastructure required to build production grade agentic applications. The primary objective of MAF is to move beyond prompt-response interactions and enable AI systems that can execute complex workflows autonomously. Developers can integrate language models, enterprise knowledge sources, APIs, databases, external services, and business processes into a unified agent architecture. The Agent Framework provides orchestration capabilities, state management, planning mechanisms, and tool integration features that simplify the development of AI agents [1].

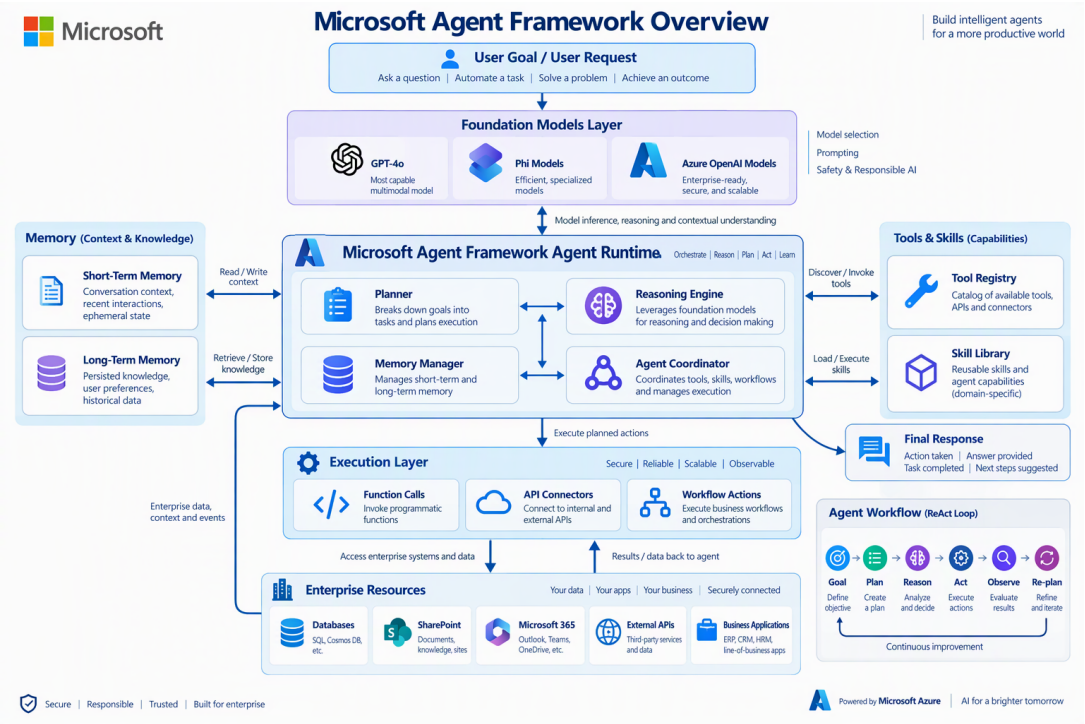


Figure 2.1: Microsoft Agent Framework Overview

2.2 Design Goals

The design principles of Microsoft Agent Framework include:

- Extensibility through pluggable tools and services
- Multi-agent collaboration capabilities
- Enterprise-grade security and governance
- Long-running workflow orchestration
- Human-in-the-loop interaction
- Integration with multiple foundation models

2.3 Core Components

MAF consists of several important components:

2.3.1 Agent

An agent serves as the primary execution unit responsible for interpreting goals and performing actions.

2.3.2 Planner

The planner decomposes complex objectives into smaller executable tasks.

2.3.3 Memory

Memory stores information from previous interactions and enables contextual decision making.

2.3.4 Tools

Tools provide access to external APIs, databases, search systems, enterprise applications, and custom business services.

2.3.5 Models

Large Language Models perform reasoning, natural language understanding, and decision generation.

2.4 Execution Workflow

The workflow generally follows the sequence:

1. Goal Input
2. Planning
3. Tool Selection
4. Execution
5. Observation
6. Response Generation

This iterative cycle enables agents to refine decisions and adapt to changing environments.

2.5 Architecture

Microsoft Agent Framework follows a layered architecture that separates reasoning, orchestration, tool execution, and memory management.

2.5.1 Agent Runtime Layer

The runtime manages agent state, task execution, event handling, and communication between components.

2.5.2 Reasoning Layer

The reasoning layer uses foundation models such as GPT-4o and other supported language models to interpret tasks and generate plans.

2.5.3 Planning Layer

The planning layer determines:

- What actions need to be performed
- Which tools should be invoked
- Whether additional information is required
- How intermediate results should be evaluated

2.5.4 Tool Integration Layer

This layer enables agents to access:

- REST APIs
- Databases
- Enterprise applications
- Search engines
- Custom functions

2.5.5 Memory Management

Short-term memory maintains conversational context, while long-term memory stores persistent knowledge and user preferences. Modern agentic systems rely heavily on memory to achieve continuity and contextual awareness.

Chapter 3

Azure AI Foundry Agent Service

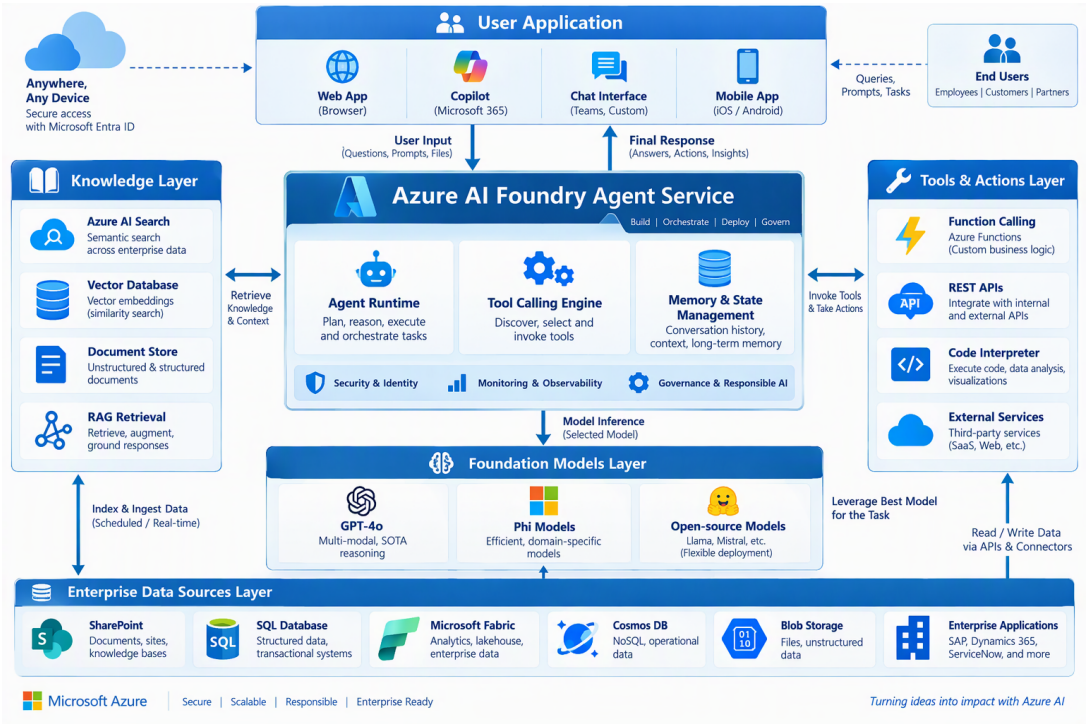


Figure 3.1: Azure AI Foundry Agent Service

3.1 Introduction

Azure AI Foundry Agent Service is a fully managed platform for building, deploying, evaluating, and governing AI agents at enterprise scale. The service extends Azure AI Foundry by providing managed infrastructure for creating agent-based applications. Developers can focus on business logic while Azure handles orchestration, scalability, monitoring, and security.

3.2 Key Features

Key capabilities include:

- Managed agent runtime
- Multi-agent orchestration
- Built-in tool calling
- Knowledge grounding using Retrieval-Augmented Generation (RAG)
- Memory management
- Evaluation pipelines
- Security and compliance controls

Azure AI Foundry Agent Service enables organizations to build agents that can retrieve information, perform actions, and integrate with enterprise systems while maintaining governance and observability [2].

3.3 Built-in Tools

The service supports:

- Bing Search
- Azure AI Search
- Code Interpreter
- Function Calling
- Knowledge Retrieval
- Custom APIs

3.4 Enterprise Integration

Organizations can connect enterprise data, documents, databases, and business systems to create domain-specific assistants with secure access control.

Chapter 4

Microsoft Agent Framework vs Azure AI Foundry Agent Service

Although both technologies support agentic AI development, they serve different purposes.

4.1 Microsoft Agent Framework

Microsoft Agent Framework primarily focuses on providing a flexible development framework and runtime environment for building custom agents.

4.2 Azure AI Foundry Agent Service

Azure AI Foundry Agent Service provides managed infrastructure for deploying and operating those agents in production environments.

4.3 Comparison

Feature	Agent Framework	Agent Service
Purpose	Development Framework	Managed Platform
Infrastructure	Self Managed	Azure Managed
Customization	High	Moderate to High
Enterprise Governance	Custom Implementation	Built-in Support
Monitoring	Developer Managed	Azure Managed
Scalability	User Responsibility	Azure Managed

Table 4.1: Comparison between Microsoft Agent Framework and Azure AI Foundry Agent Service

Bibliography

[1] URL: https://learn.microsoft.com/en-us/agent-framework/.

[2] URL: https://learn.microsoft.com/en-us/azure/ai-foundry/agents/.